



Mohamed Aziz Skhiri

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Profile | Data Scientist / Full-Stack Developer with hands-on experience building end-to-end ML pipelines, predictive models, and production systems. Reduced data processing time by 35%; pipelines handling over 10,000 records/day. Full-stack aircraft fuel prediction platform (XGBoost, LSTM, RAG) deployed on the cloud, in partnership with Nouvelair Tunisie. Proficient in Python, SQL, Flask, React, AWS.

Key Skills

- **Programming & Data:** Python, R, SQL, Java, C++, C#.NET, Bash, Git, Linux
- **Machine Learning:** XGBoost, LSTM, Regression, Classification, Clustering, Hyperparameter Tuning, SHAP, Model Evaluation, TensorFlow/Keras, Scikit-learn
- **Data Engineering & MLOps:** ETL Pipelines, REST APIs,
- **Visualization & BI:** Power BI, Tableau, Matplotlib, Seaborn, React/Recharts Dashboards
- **Full-Stack Development:** React, Flask, FastAPI, JWT, RBAC, REST APIs

Professional Experience

Freelance Data Scientist / Developer

2024 – Present

Independent / Upwork

- Designed and deployed **MedRAG**, a RAG-based drug information assistant: **vector database** (embedding indexing, cosine similarity search) combined with an LLM to query drug leaflets for indications, precautions and interactions; built and deployed the ingestion/vectorization pipeline on **Linux/Ubuntu**.
- Built a **Global Health Data Engineering** pipeline, cleaning and unifying global health indicators into an interactive **Streamlit** dashboard with choropleth maps and dynamic filters.

Data Scientist / Full-Stack Developer – Graduation Internship (PFE)

Feb 2026 – Jun 2026

Nouvelair Tunisie, IT & Projects Technology Department

- Designed FuelTrack, an ML platform for monitoring, predicting and optimizing aircraft fuel consumption and CO₂ emissions, replacing an entirely manual workflow based on LIDO Flight, PDC FlightOps and OFP Excel files.
- Developed and validated predictive models (XGBoost, Gradient Boosting, Random Forest, LSTM) to estimate Block Fuel, Trip Fuel, costs and CO₂ ($R^2 > 99\%$), with per-registration bias correction (HFC) to improve accuracy.
- Designed a RAG-based conversational assistant (Groq LLaMA 3.3-70B, with Gemini 2.5 Flash fallback) enabling natural-language querying of aggregated flight data in SQL.
- Delivered a production-grade full-stack architecture (Flask, PostgreSQL, React) with JWT authentication and 4-tier role management, using a hybrid SCRUM + CRISP-DM methodology across 6 sprints, under academic and professional supervision.

Full-Stack Developer Intern – Data/AI Module

May 2025 – Aug 2025

Elite Council Consulting

- Built backend pipelines for an AI-driven attendance management system processing over 5,000 events/day.
- Designed and deployed React dashboards for real-time analytics, improving reporting efficiency by 40%.
- Refactored Flask/MongoDB modules, improving data flow reliability by 25%.

Python Programming Instructor

Feb 2024 – Jun 2024

CEFI – Computer Training Center

- Trained over 30 students in Python and data structures through hands-on analysis mini-projects; designed practical exercises and assignments on data manipulation.

Key Projects

FuelTrack – Intelligent Aircraft Fuel Prediction & Recommendation Platform

- **Prediction** : XGBoost, LSTM, Random Forest, Gradient Boosting models estimating Block Fuel, Trip Fuel and CO₂ ($R^2 > 99\%$), with SHAP explainability and per-registration bias correction.
- **Route recommendation** : intelligent engine (Route Recommender & Route Decoder) suggesting the most fuel-efficient routes based on aircraft registration, with dynamic filters and real-time flight-count badges.
- **Dashboards & analytics** : Power BI dashboards and Python analytics (Pandas, Matplotlib/Seaborn) with anomaly detection (Z-score) and fuel savings tracking.
- **Chatbot** : RAG-based conversational assistant (Groq LLaMA 3.3-70B) enabling natural-language queries over flight data aggregated in SQL.
- Production-grade full-stack architecture : Flask, PostgreSQL, React, JWT authentication and 4-tier RBAC.

Other Projects

- **Football Data Pipeline & Analytics Dashboard**: ETL pipeline processing over 20,000 matches with predictive feature engineering; Flask + Docker API exposing statistics through interactive dashboards.
- **Smart School Attendance System**: ML backend processing over 5,000 events/day with real-time React dashboards (Flask, MongoDB, Docker).

Source code for all projects available at github.com/skhiri23

Education

Professional Master's in Data Science – Faculty of Sciences of Monastir (FSM), Univ. of Monastir

2024–2026

- Intensive training in Big Data, Machine Learning and AI: Introduction to Big Data, SQL/NoSQL Databases, Supervised & Unsupervised Learning, Parallel & Distributed Computing, Distributed Systems, Artificial Intelligence, Natural Language Processing (NLP), Big Data Frameworks, Data Engineering and Advanced Python Programming.
- *Graduation project (PFE)*: FuelTrack – design and development of a full-stack ML platform for aircraft fuel monitoring and optimization, in partnership with Nouvelair Tunisie & ATDS (see Projects section).

Bachelor's in Software Engineering & Information Systems – Faculty of Sciences of Monastir (FSM), Univ. of Monastir**2021–2024**

- Core training in software engineering and information systems: Algorithms & Data Structures, OOP (Java), Operating Systems & Networks, Databases, Web Technologies, Software Engineering, SOA Architecture & Web Services, Machine Learning, Information Security, Virtualization & Cloud, Mobile & Distributed Application Development.
- *Graduation project (PFE)*: Human Resources Management System (HRMS) .

Certifications

- Python for Data Science, AI & Development – IBM
- AWS Cloud Practitioner (Fundamentals) – AWS
- Full-Stack Web Development (React, Flask, MongoDB) – ECC
- Python Programming Certification – JNL Center

Languages

- **Arabic**: Native language
- **English**: Professional level
- **French**: Intermediate level